

EYE-CARE: Eye Tracking for Context-Adaptive Systems Based on Activity and Cognitive Load Prediction

Kenan Bektaş
kenan.bektas@unisg.ch
University of St. Gallen
St.Gallen, Switzerland

Amine Abbad-Andaloussi
amine.abbad-andaloussi@unisg.ch
University of St. Gallen
St. Gallen, Switzerland

Abstract

With the growing amount of digital data, the importance of systems that allow humans to exploit such data to obtain *insightful interpretations* has skyrocketed. Recent context-adaptive systems can provide users with assistance by proactively sensing, tracking, and interacting with them in a seamless, user friendly, and privacy preserving manner. For example, in an industrial or educational setting, such a system can detect when a user is challenged and provide adequate support (e.g., by highlighting task-relevant information that remains beyond the user's attention). To maintain such adaptations, a real-time assessment of the user's cognitive state and behavior are required. Such assessments can be addressed in multiple ways. One particular way to understand user behavior is to assess their visual perception through gaze-enabled systems. This half day tutorial will enable participants to *analyze* an existing human-activity recognition pipeline that is trained with an exiting eye movement dataset. The participants will be encouraged to *build* groups and *discuss* how this pipeline (as a proxy environment) can be extended (e.g., with multi-modal interaction or large language models) towards providing users with feedback that is potentially useful in a broad spectrum of personal and professional activities.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; Mixed / augmented reality; • **Computing methodologies** → Perception; Supervised learning.

Keywords

Eye tracking, human activity recognition, context-adaptive systems, workload, tutorial

ACM Reference Format:

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1 Introduction

Benefits. Eye tracking data can be collected through conventional desktop or mobile systems or more contemporary technologies such as mixed reality headsets, smart glasses, or dynamic vision sensors. Eye tracking can provide insights into user attention, intentions, and activities, and allow developing novel interaction methods in ubiquitous computing environments. In this half day tutorial, participants will be introduced to human activity and cognitive load prediction by means of eye tracking. Furthermore, we will discuss technical and privacy related concerns related to collecting eye tracking data and more specific challenges in predicting human activities and cognitive load from eye movements. Participants will analyze machine learning pipelines (i.e., Python notebooks accompanies with eye movement datasets) that are trained in predicting human activities and cognitive load. Then they will be encouraged to build groups and discuss how such pipelines can be improved and extended towards providing users with feedback that can be contextually relevant in their personal or professional activities. By the end of this tutorial, we expect that the covered material and group discussions will help participants to build fruitful scientific collaborations among themselves.

Intended Audience. The tutorial is primarily targeted at graduate students, researchers, and industrial practitioners in human-computer interaction, ubiquitous computing, and software engineering with specific interest in developing (mobile) eye tracking applications that can augment perceptual and cognitive abilities of humans [3, 5].

Prerequisites. The audience is expected to be motivated in (ubiquitous) applications of eye tracking and have a preliminary understanding of machine learning techniques. In this tutorial, we will provide a brief introduction to these topics. Furthermore, Python notebooks and eye tracking datasets will be provided in advance to allow participants to interactively learn, during the tutorial, to train models for human activity recognition, which will also provide a basis for them to develop new models later. Therefore, participants of this tutorial are expected to have an introductory level of experience in programming with Python ¹.

2 Content

This tutorial aims to provide an introductory training for developing context-adaptive systems and involves the following parts:

- Overview of eye-tracking data collection protocols and extraction of key features allowing to infer users' activities,

¹<https://www.python.org/>

cognitive load and behavioral processes. Notably, we will cover the following aspects:

- Introduction to eye tracking for novices
- Overview of eye-tracking data collection protocols
- Raw data processing pipelines (oculomotor event detection, pupil cleaning)
- Key eye-tracking concepts (fixations, saccades, AOI dwell, scan paths)
- Eye-tracking metrics and features reflecting cognitive load, behavioral processes, and user activities
- Using machine learning to predict user activities (cf. [3]), cognitive load (cf. [2]), and behavioral processes (cf. [1, 4]). We will cover the following aspects:
 - Machine learning approaches that can be used in predicting user activities, cognitive load, and behavioral processes
 - Common challenges in predicting user activities, cognitive load, and behavioral processes
 - In the practical part, participants will have access to Python notebooks to interact with some provided machine learning models and eye-tracking data sets.
- Designing context-specific adaptive interventions (e.g., visual cues, pacing, feedback). In particular, we will cover the following aspects:
 - Overview and examples of context-specific interventions.
 - Collaborative group activity
 - * Participants select a topic relevant to their area of expertise (e.g., ubiquitous computing or software engineering)
 - * Define possible interventions triggered by detected cognitive load or inferred cognitive processes or activities (e.g., provide real-time explanations of the parts of code the user is struggling with)
 - * Design how interventions are presented (e.g., a side pane in an IDE providing contextual explanations)
 - * Discuss how large language models (e.g., OpenAI GPT 5²) can support intervention delivery
 - * Provide access to the OpenAI API and ask participants to implement a simple intervention

3 Instructor Background

Dr. Kenan Bektaş. Kenan is a postdoc researcher and lecturer in the Interaction- and Communication-based Systems Group of the Institute of Computer Science, the University of St. Gallen, Switzerland and also works as a senior researcher in the Cantonal Hospital St. Gallen. He received his Ph.D. degree from the University of Zurich, Faculty of Science, in 2018. Kenan's research interests comprise human visual perception, embodied human-computer interaction, human augmentation, and pervasive applications of eye tracking. He coauthored multiple publications and reviewed papers in venues such as the ACM ETRA, CHI, IMWUT, IUI, DIS and the IEEE TVCG.

Dr. Amine Abbad-Andaloussi. Amine is a postdoctoral researcher and lecturer in the Software Systems Programming and Development Group of the Institute of Computer Science, the University

of St. Gallen, Switzerland. He received his Ph.D. degree from the Technical University of Denmark, Department of Applied Mathematics and Computer Science in 2021. Amine's research focuses on the human and cognitive aspects of software and process engineering. He's a member of several research communities (as co-author, reviewer), including BPM, ICPM, PSJ, IS, JSS, and ESWA.

Both instructors teach eye-tracking methods as part of different courses, including "Interactive Information Visualization - From Theory to Experiments" and "Software Assessment", both organized at the University of St Gallen.

4 Resources

The tutorial will be based on our research on the topic [1–4]. Tutorial-specific materials (e.g., a set of slides, an eye-tracking dataset, and Python notebooks) are stored in <https://github.com/Interactions-HSG/ETRA26-EYECARE-Tutorial> which is adapted from <https://github.com/Interactions-HSG/GEAR>.

5 Accessibility

In this tutorial, participants will have access to Python notebooks, which will offer tools such as the Accessibility Toolbar (AT4N³) that will be helpful for those who are who are dyslexic or visually impaired.

References

- [1] Amine Abbad-Andaloussi, Clemens Schreiber, and Barbara Weber. 2024. Using Eye-Tracking to Detect Search and Inference During Process Model Comprehension. In *Cooperative Information Systems*, Marco Comuzzi, Daniela Grigori, Mohamed Sellami, and Zhangbing Zhou (Eds.). Springer Nature Switzerland, Cham, 21–38.
- [2] Amine Abbad-Andaloussi, Thierry Sorg, and Barbara Weber. 2022. Estimating developers' cognitive load at a fine-grained level using eye-tracking measures. In *Proceedings of the 30th IEEE/ACM International Conference on Program Comprehension (Virtual Event) (ICPC '22)*. Association for Computing Machinery, New York, NY, USA, 111–121. doi:10.1145/3524610.3527890
- [3] Kenan Bektaş, Jannis Strecker, Simon Mayer, and Kimberly Garcia. 2024. Gaze-enabled activity recognition for augmented reality feedback. *Computers & Graphics* (2024), 103909. doi:10.1016/j.cag.2024.103909
- [4] Damian Hostettler, Kenan Bektaş, and Simon Mayer. 2023. Pupillometry for Measuring User Response to Movement of an Industrial Robot. In *Proceedings of the 2023 Symposium on Eye Tracking Research and Applications* (Tubingen, Germany) (ETRA '23). Association for Computing Machinery, New York, NY, USA, Article 52, 2 pages. doi:10.1145/3588015.3590123
- [5] Roope Raisamo, Ismo Rakkolainen, Päivi Majaranta, Katri Salminen, Jussi Rantala, and Ahmed Farooq. 2019. Human augmentation: Past, present and future. *International Journal of Human-Computer Studies* 131 (Nov. 2019), 131–143. doi:10.1016/j.ijhcs.2019.05.008

²<https://openai.com/gpt-5/>

³<https://pypi.org/project/accessibility-toolbar/>